# Final Project for Genomic Data Analysis

Yihao Zhao

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## 1 Introduction

The GSE11877 data set (pediatric ALL) from the Children's Oncology Group P9906 study was used, which contains information on 207 high-risk pediatric B-cell acute lymphoblastic leukemia (ALL) patients profiled at diagnosis using the Affymetrix HG-U133 Plus 2.0 array. The data set includes raw microarray data and several baseline clinical variables, such as age, sex, race/ethnicity, and white blood cell (WBC) count at diagnosis. The WBC count at diagnosis is a standard clinical risk factor in pediatric ALL. The commonly used cutoff of 50,000/µL has long been part of the NCI/COG risk-classification system, in which higher levels generally reflect more aggressive disease biology (Smith et al., 1996).

Therefore, patients were divided into high WBC ($\geq$ 50,000/µL) and low WBC ($<$ 50,000/µL) groups, and focus on the question: **Do children with high WBC counts at diagnosis exhibit different gene expression patterns compared to those with lower counts, and which genes contribute the most to this difference?**

## 2 Methods

### Pre-processing

The raw CEL files for GSE11877 were downloaded and used to created an Affymetrix `AffyBatch` object. All arrays were pre-processed using the RMA method, which includes background correction, quantile normalization, $\log_2$ transformation, and probe-set summarization.

After normalization, two standard filtering steps were applied to remove non-informative probes. First, probe sets with mean $\log_2$ expression $\leq 4$ were removed, since Affymetrix intensities below this range generally behave like background noise and do not reflect reliable biological signal. Second, probes with variance below the median were removed, because genes with very low variability across samples do not contribute meaningfully to PCA or clustering and tend to dilute statistical power.

After filtering, the dataset contained approximately 9,700 probe sets and 207 samples. The processed expression matrix and phenotype data (including the new `wbc_group` variable) were stored together in a final `ExpressionSet` object for reproducible downstream analyses.

## Dimensional Reduction

To explore major expression patterns, principal component analysis (PCA) was performed on the filtered, RMA-normalized expression matrix. Since the dataset includes thousands of genes measured on different scales, `prcomp()` was used with `scale = TRUE` so that each gene contributed equally. PCA was applied to the samples (columns), and the proportion of variance explained by each principal component was extracted. A scree plot and a PC1 vs. PC2 scatterplot were generated to examine whether samples separated by WBC group (High vs. Low).

## Clustering

Hierarchical clustering was first performed to examine whether natural subgroups appeared based on global gene-expression patterns. Using the filtered RMA-normalized matrix, Euclidean distances between samples were computed, and agglomerative hierarchical clustering with complete linkage was applied. The resulting dendrogram was visualized, and cluster membership was evaluated using a two-cluster cut.

As a second unsupervised method, K-means clustering was applied to the same dataset. Because K-means is sensitive to gene scale, the expression matrix was standardized by centering and scaling each gene before clustering. K-means was run with $K = 2$ and `nstart = 25` to ensure a stable solution. To assess consistency between the two clustering approaches, a contingency table was constructed comparing hierarchical and K-means cluster assignments. The K-means clusters were also visualized in PCA space.

### Differential Expression

Differential expression analysis was performed using the `limma` workflow on the filtered, RMA-normalized matrix. WBC group was defined as a two-level factor (Low vs. High), and a design matrix without an intercept was constructed. Gene-wise linear models were fit using `lmFit`, and a contrast comparing HighWBC − LowWBC was created. Empirical Bayes moderation (`eBayes`) was applied to stabilize variance estimates across probes.

Gene-level results were extracted using `topTable`, and p-values were adjusted using the Benjamini–Hochberg FDR procedure. Genes were annotated with the `hgu133plus2.db` package to obtain gene symbols and Entrez IDs. Genes with FDR $< 0.05$ were considered statistically significant, and the top 10 differentially expressed genes were summarized.

### Supervised Learning

For supervised classification, a penalized logistic regression model was fit using all filtered genes as predictors and WBC group as the binary outcome (Low WBC = 0, High WBC = 1). For each value of $\lambda$ along the lasso path, 10-fold cross-validation was performed, and the AUC was calculated based on predicted probabilities. The value of $\lambda$ that produced the highest AUC was selected, and the corresponding coefficient vector was extracted. The number of non-zero coefficients in this model represents the genes selected by the lasso.

A random forest classifier was also trained with 5,000 trees, using the transposed expression matrix (samples $\times$ genes) as predictors and WBC group as the outcome. For each gene, variable-importance scores (MeanDecreaseGini) were calculated and used to rank genes. The top 10 genes with the highest importance scores were reported.

## 3 Results

### Pre-processing

The raw probe-intensity boxplot showed large differences in overall intensity levels across arrays, with misaligned medians and uneven spreads. As shown in Figure 1, the raw intensities varied widely across samples, suggesting substantial technical variation that required normalization.

After applying RMA, the distributions became tightly aligned across all 207 samples (Figure 2), indicating that quantile normalization successfully removed between-array variation. Following normalization, probe sets with very low expression or very low variance were filtered out, reducing the dataset from 54,675 total probe sets to approximately 9,756 before downstream analysis.

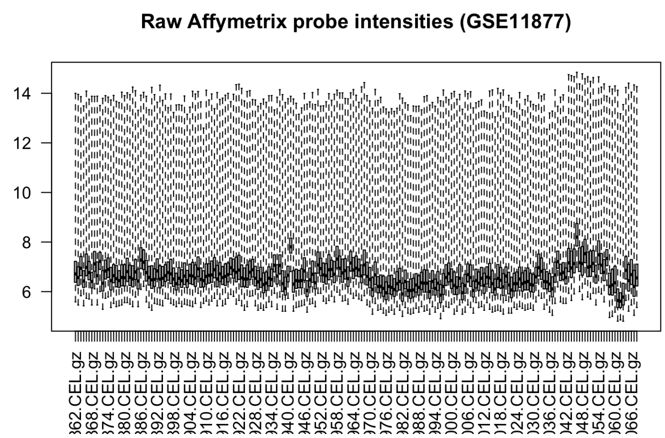


Figure 1: Raw Affymetrix probe intensities across 207 samples before normalization.

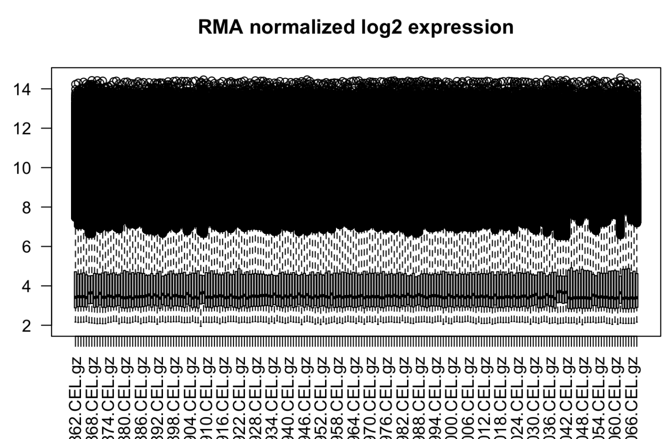


Figure 2: RMA-normalized $\log_2$ expression values showing aligned distributions across samples.

## Dimensional Reduction

The scree plot (Figure 3) showed that PC1 and PC2 explained about 24% and 12% of the variance. In the PC1–PC2 scatterplot (Figure 4), high-WBC and low-WBC samples overlapped heavily, suggesting that WBC level does not drive major global expression differences. A small shift along PC2 indicated that some more subtle gene-level differences may still exist.

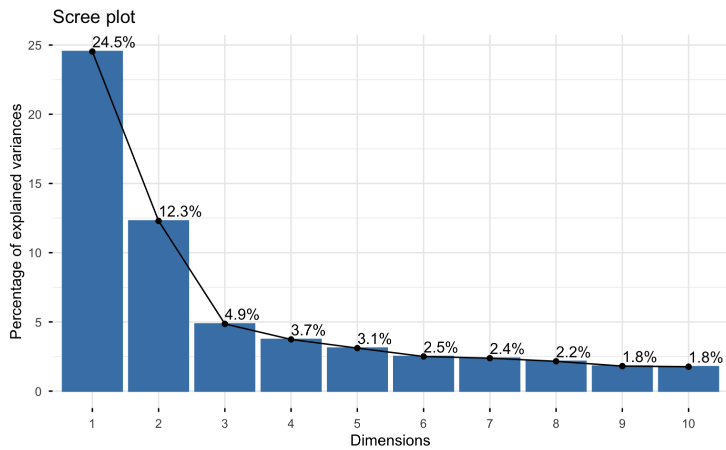


Figure 3: Scree plot showing variance explained by the top principal components.

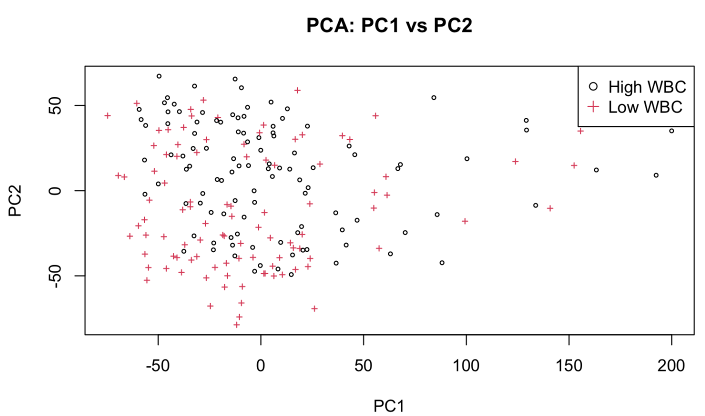


Figure 4: PC1 vs. PC2 scatterplot colored by WBC group (High vs. Low).

## Clustering

Hierarchical clustering (Figure 5) produced one dominant cluster and a much smaller secondary cluster, with no evidence of separation by WBC group. The dendrogram showed that most samples had similar expression patterns, and the heatmap (Figure 7) did not form clear block structures.

The K-means solution with $K = 2$ showed the same pattern: one large cluster and one small cluster. Cluster assignments were almost identical between hierarchical clustering and K-means. Visualizing K-means clusters in PCA space (Figure 8) again showed heavy overlap, with no clear expression-based subgroups.

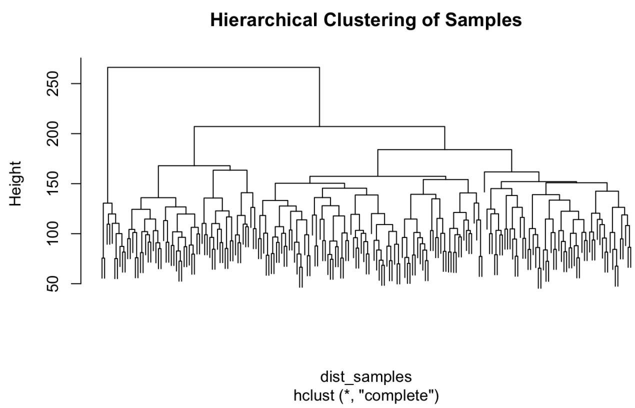


Figure 5: Hierarchical clustering dendrogram with complete linkage.

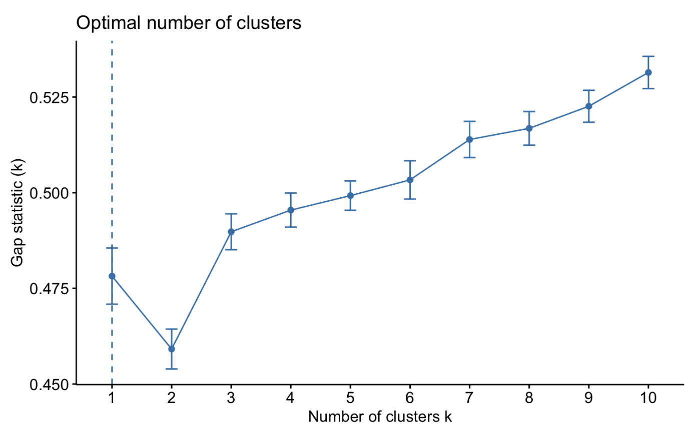


Figure 6: Gap statistic for $k = 1$–$10$ clusters. The optimal value occurs at $k = 1$, indicating no strong evidence for multiple natural clusters in the data.

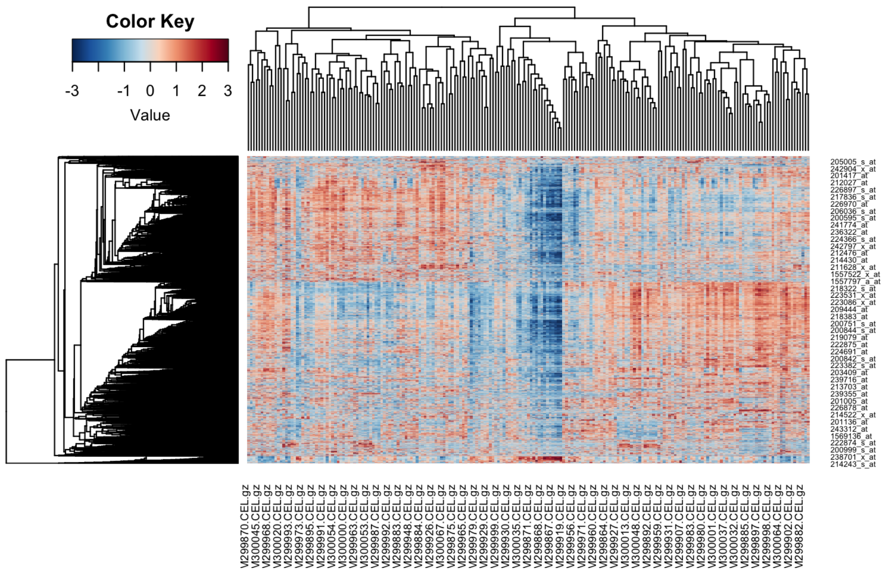


Figure 7: Hierarchical clustering heatmap with sample dendrogram (complete linkage). Rows represent genes and columns represent samples. No clear separation between high and low WBC groups is visible.

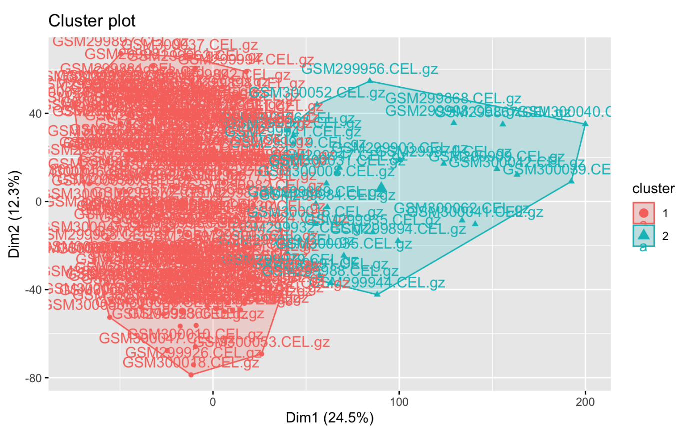


Figure 8: K-means clustering results visualized in PCA space.

## Differential Expression

Using limma, 2,650 genes with significant expression differences between High WBC and Low WBC groups were identified(FDR $< 0.05$). The top genes included *CRIP1*, *NARF*, *MED1*, *GCC1*, *PSMD12*, *KLF10*, *METRNL*, *RANBP2*, *FUBP3*, and *FEM1B*. These genes (Table 1) showed the largest expression differences and the smallest adjusted p-values.

| Gene Symbol | Low WBC | High WBC | P-value | FDR |
|---|---|---|---|---|
| CRIP1 | 5.222 | 6.679 | $1.68 \times 10^{-12}$ | $1.63 \times 10^{-08}$ |
| NARF | 4.998 | 4.244 | $1.35 \times 10^{-10}$ | $6.60 \times 10^{-07}$ |
| MED1 | 5.322 | 4.608 | $3.74 \times 10^{-10}$ | $1.22 \times 10^{-06}$ |
| GCC1 | 5.830 | 5.280 | $1.74 \times 10^{-09}$ | $4.25 \times 10^{-06}$ |
| PSMD12 | 4.812 | 4.115 | $2.54 \times 10^{-09}$ | $4.69 \times 10^{-06}$ |
| KLF10 | 8.899 | 7.852 | $5.29 \times 10^{-09}$ | $4.69 \times 10^{-06}$ |
| METRNL | 5.350 | 4.446 | $5.13 \times 10^{-09}$ | $4.69 \times 10^{-06}$ |
| RANBP2 | 4.861 | 4.084 | $3.87 \times 10^{-09}$ | $4.69 \times 10^{-06}$ |
| FUBP3 | 5.599 | 4.964 | $3.71 \times 10^{-09}$ | $4.69 \times 10^{-06}$ |
| FEM1B | 4.530 | 3.938 | $6.66 \times 10^{-09}$ | $5.42 \times 10^{-06}$ |

Table 1: Top 10 differentially expressed genes between High and Low WBC groups (FDR $< 0.05$).

## Supervised Learning

The lasso logistic regression model achieved a maximum AUC of about 0.90, showing strong ability to classify High vs. Low WBC samples. At the optimal $\lambda$, the lasso retained 69 non-zero coefficients out of roughly 9,700 genes, suggesting that a relatively small gene set carries most of the predictive signal.

The random forest classifier also showed good predictive performance. Using 5,000 trees, it ranked genes by variable importance. The top 10 genes (Table 2) were *CRIP1*, *MED1*, *SURF4*, *DDX3Y*, *TAGLN2*, *TMEM87A*, *GNLY*, *FBXO34*, *TENT5C*, and *CCDC93*. These genes contributed the most to distinguishing High vs. Low WBC groups.

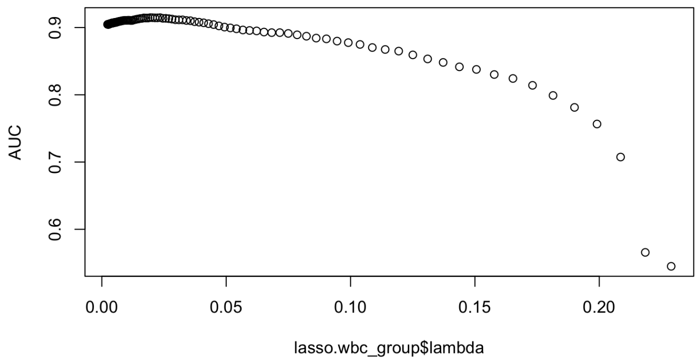


Figure 9: Cross-validated AUC across the lasso regularization path.

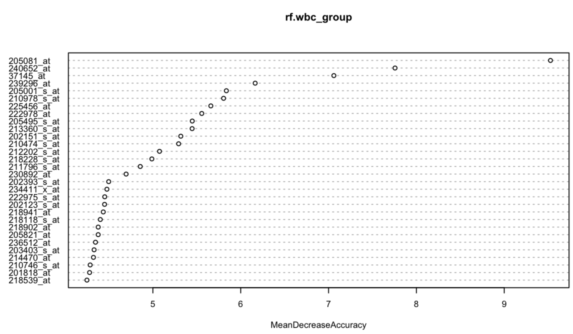


Figure 10: Random forest variable-importance plot (MeanDecreaseGini).

| Gene ID | Gene Symbol | MeanDecreaseGini |
|---|---|---|
| 205081_at | CRIP1 | 0.521 |
| 225456_at | MED1 | 0.245 |
| 222978_at | SURF4 | 0.218 |
| 205001_s_at | DDX3Y | 0.211 |
| 210978_s_at | TAGLN2 | 0.185 |
| 212202_s_at | TMEM87A | 0.171 |
| 37145_at | GNLY | 0.169 |
| 218539_at | FBXO34 | 0.162 |
| 226811_at | TENT5C | 0.162 |
| 209688_s_at | CCDC93 | 0.158 |

Table 2: Top 10 genes ranked by Random Forest variable importance (MeanDecreaseGini).

# 4 Discussion

In this project, the goal was to examine whether children with high WBC counts at diagnosis exhibit different gene-expression patterns compared to those with lower counts, and to identify which genes contribute most to this difference. Overall, the unsupervised results suggested that WBC level does not drive broad expression changes. Both PCA and clustering showed substantial overlap between the two groups, with no strong global structure separating high and low WBC patients. This indicates that overall transcriptional profiles are quite similar, and that any differences are more likely gene-specific rather than global.

In contrast, the gene-level analyses told a different story. Differential expression analysis identified 2,650 significant genes, indicating that many individual genes change between the high and low WBC groups. The supervised models supported this finding. The lasso model achieved an AUC of approximately 0.90 and required only 69 genes for accurate classification. The random forest model also performed well and consistently highlighted important genes such as *CRIP1* and *MED1*. The recurrence of these genes across methods strengthens their potential role in biological differences related to WBC level.

In conclusion, these results suggest that WBC count is not associated with large-scale shifts in gene-expression patterns, but is linked to more focused gene-level changes.

# References

1. Smith M. *et al.* (1996). Uniform approach to risk classification and treatment assignment for children with acute lymphoblastic leukemia. *New*

*England Journal of Medicine*.

2. Kang H., Chen I.M., Wilson C.S., Bedrick E.J. *et al.* (2010). Gene expression classifiers for relapse-free survival and minimal residual disease improve risk classification and outcome prediction in pediatric B-precursor acute lymphoblastic leukemia. *Blood*, 115(7), 1394–1405.

3. Harvey R.C., Mullighan C.G., Wang X., Dobbin K.K. *et al.* (2010). Identification of novel cluster groups in pediatric high-risk B-precursor acute lymphoblastic leukemia with gene expression profiling: correlation with genome-wide DNA copy number alterations, clinical characteristics, and outcome. *Blood*, 116(23), 4874–4884.

4. Harvey R.C., Mullighan C.G., Chen I.M., Wharton W. *et al.* (2010). Rearrangement of CRLF2 is associated with mutation of JAK kinases, alteration of IKZF1, Hispanic/Latino ethnicity, and poor outcome in pediatric B-progenitor acute lymphoblastic leukemia. *Blood*, 115(26), 5312–5321.